

Hydrocarbons

THE DIFFERENT FAMILIES OF COMPOUNDS MADE PURELY FROM CARBON AND HYDROGEN.

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Hydrocarbons are the simplest compounds in living things. The study of chemicals found in living things is called organic chemistry.

Hydrocarbon chains

Carbon atoms can form up to four covalent bonds. This allows carbon to form intricate hydrocarbon molecules. The carbon atoms are chained together, with hydrogen atoms bonded to the spare electrons. When hydrogen atoms are not available, two carbon atoms may form double, and even triple, bonds.

▷ **Chained molecule**
This is the hydrocarbon octane (C_8H_{18}).

carbon atom bonded to three hydrogen atoms and one carbon atom

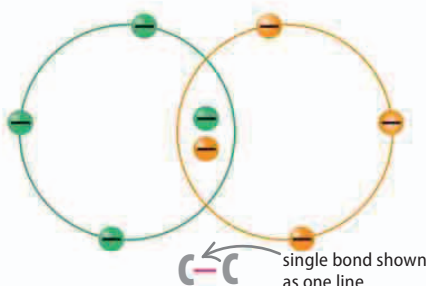
carbon atom bonded to two hydrogen atoms and two carbon atoms

carbon atom

hydrogen atom

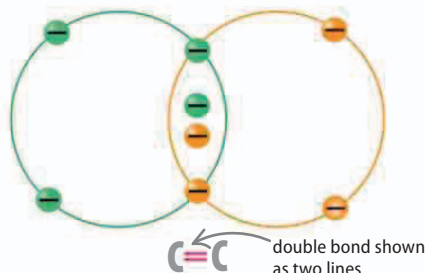


◁ **Letter diagram**
This is octane shown simply with the chemical symbols for carbon (C) and hydrogen (H).



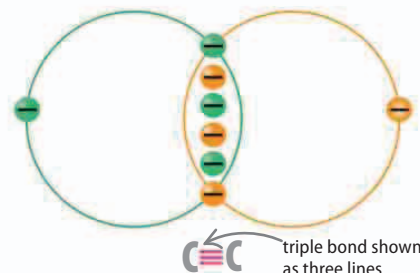
△ Single bond

The normal carbon-to-carbon covalent bond involves sharing a single pair of electrons.



△ Double bond

This bond has two pairs of electrons shared between the atoms. It is less stable than a single bond.



△ Triple bond

This very unstable bond contains three shared pairs of electrons to form a triple bond.

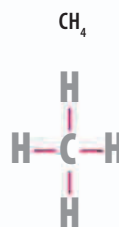
Naming system

Hydrocarbons with chained and branched molecules are known as aliphatics. They are named with a prefix that is specific to the number of carbon atoms in their longest chain. Side branches on the main chain are also named using the same prefixes. These branches are known as alkyl groups, and so the prefix is followed by the suffix “-yl” to show they relate to a branch other than the main chain. For example a methyl chain is a side branch with one carbon atom.

Prefix	Number of carbon atoms
meth	1
eth	2
prop	3
but	4
pent	5
hex	6

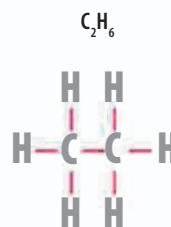
△ Prefixes

The first four prefixes are specific to hydrocarbons, while from five onward the prefixes are based on Latin and Greek numbers.



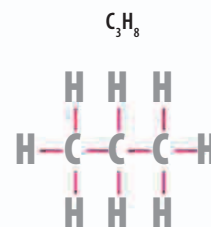
△ Methane

The simplest hydrocarbon is also known as natural gas and is burned as a fuel.



△ Ethane

Ethane has two carbon atoms, and is the compound used to make polythene plastic.



△ Propane

With three carbon atoms, propane gas is the fuel supplied in the tanks used in camping stoves.